

Introduction to Robotics

Spring 2022

Course Project's Handout

Version 2.

Last updated: June 9th, 2022

Summary of Changes

- The following changes were made to encourage everyone who have suffered by the lidar sensor and to relief unnecessarily competitive atmosphere. Check further details on relevant pages.
- Scoring rule (p.11), Task scores (pp.12-15)
 - Task score solely considers the completeness of subtasks
 - Task score range is scaled down from 100 to 7.5 as it is in the overall score ($7.5 \times 4 = 30$)
 - In each task, the top 3 teams get additional 1.5 points
 - For Task 2, 3, and 4, we provide arena maps on BB>Contents (How to use it without interacting with rviz: Check the code of Milestone 1 and ROS Wiki)
- Additional ground rules during evaluations (p.16)
- Evaluation schedule (p.17) and postscript (p.18)

Room Reservation

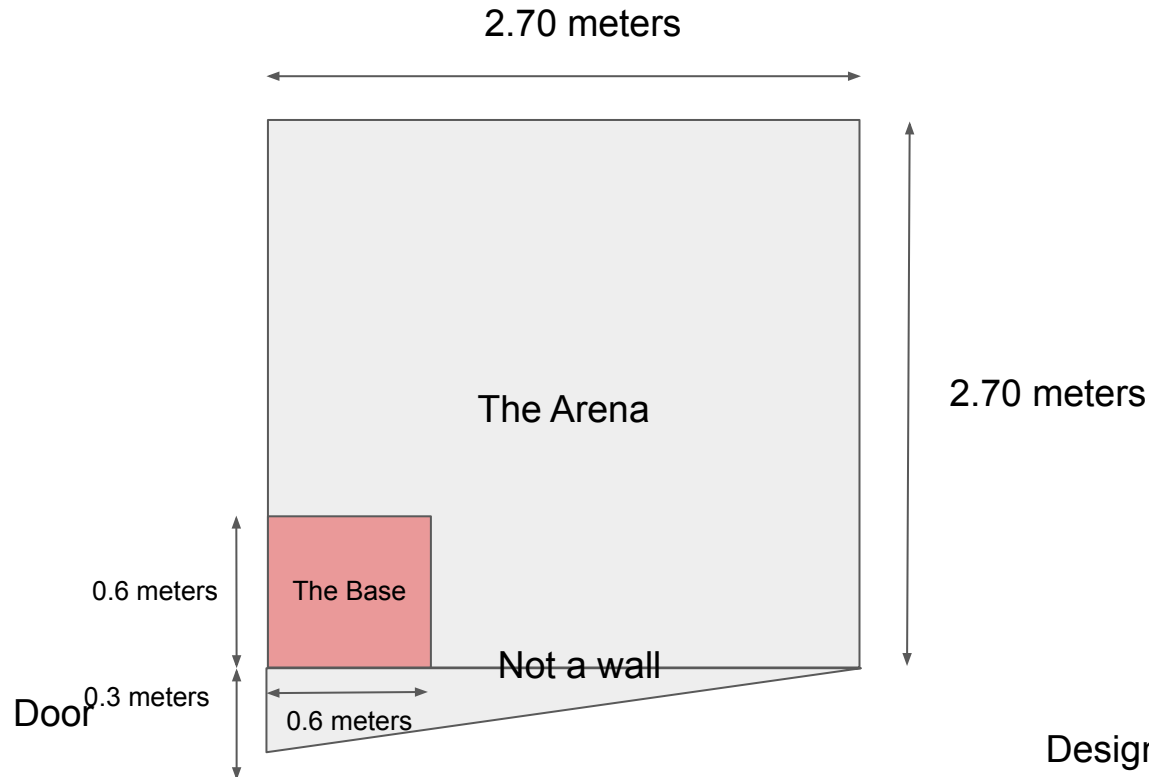
- We have reserved the same room for this course project
 - Please reserve the room in advance to give your team chance to test your code on a real hardware: <https://doodle.com/meeting/participate/id/b2kkVjJd>
- In the room, you can find
 - The TurtleBot
 - Two small bottles of different weight for testing the robot arm's picking skill
 - An arena in which the TurtleBot will be evaluated.
 - Center chamber and pressure pad for tasks
 - A portable monitor and a keyboard/mouse combo.
 - A portable wifi router
 - etc.
- After using the room, please put the items nicely for the next team.
- If you notice that some items are broken, you must report to us asap.

Tasks

- The course project consists of five tasks:
 - » Task 1: Approach a bottle and pick it up (details on p.12)
 - » Task 2: Bring all bottles back to the base in an arena without walls (details on p.13)
 - » Task 3: Remove a bottle from a chamber (details on p.14)
 - » Task 4: Use a bottle to activate a chamber's door to remove another bottle from the chamber (details on p.15)
- You can write one program for the five tasks, or write five programs for each task.
 - » When you start your program, your program can accept one argument to indicate which task it is
 - But it should be the only argument your program accepts.
- Your robot has to be fully autonomous
 - » After you start your program for a task, you cannot remotely interact with your program.
 - » Except for the emergency situation in which you need to stop the robot.
 - » You **may not interact with gui** such as rviz or robotis gui for arm **during the attempts.**
- All attempts will be audio and video recorded for post analysis that might be needed.

The Arena

- The dimension of the arena is around 2.70 meters x 2.70 meters.
- There are walls surrounding the arena. The turtlebot must stay inside the arena at all times.
- The base is the 0.6 meters x 0.6 meters region at the lower left corner of the arena.
- For Task 3 and Task 4, a center chamber will be added in the arena.
- For Task 2 and Task 4, a pressure pad will be added in the arena.



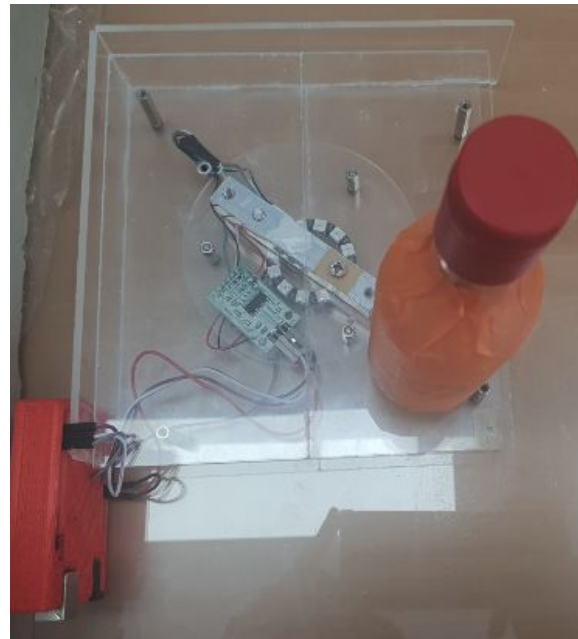
The Arena with center chamber location (cont.)



Constructed by Minhyuk Park
& Jaeyeon Lee

Two Bottles

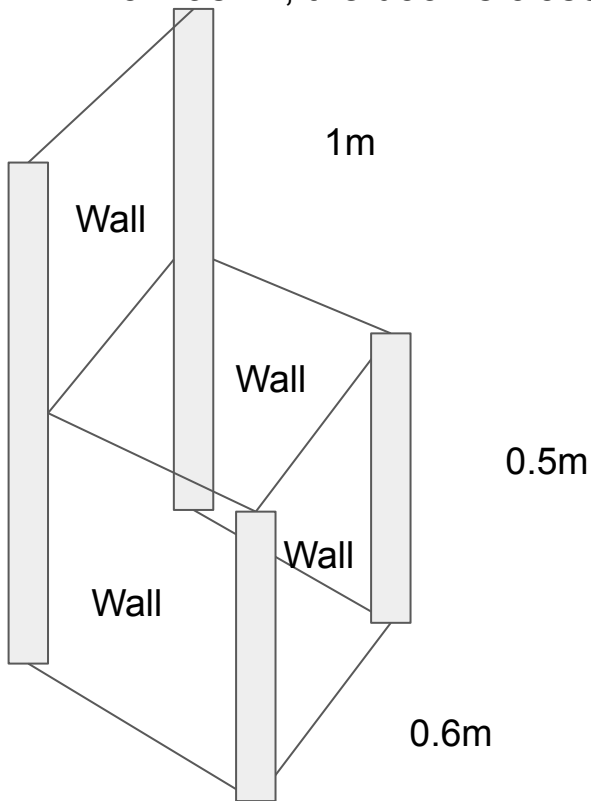
- There are only two bottles, each with different weight (Bottle 1: Lighter, Bottle 2: Heavier)
- You can use the **default YOLO v3 tiny model** to recognize the bottles.
- Sometimes YOLO misclassifies the bottle. But the bounding boxes of the misclassified objects are still useful for locating and grasping the bottles.



Light(1) and heavy(2) bottles

Center chamber

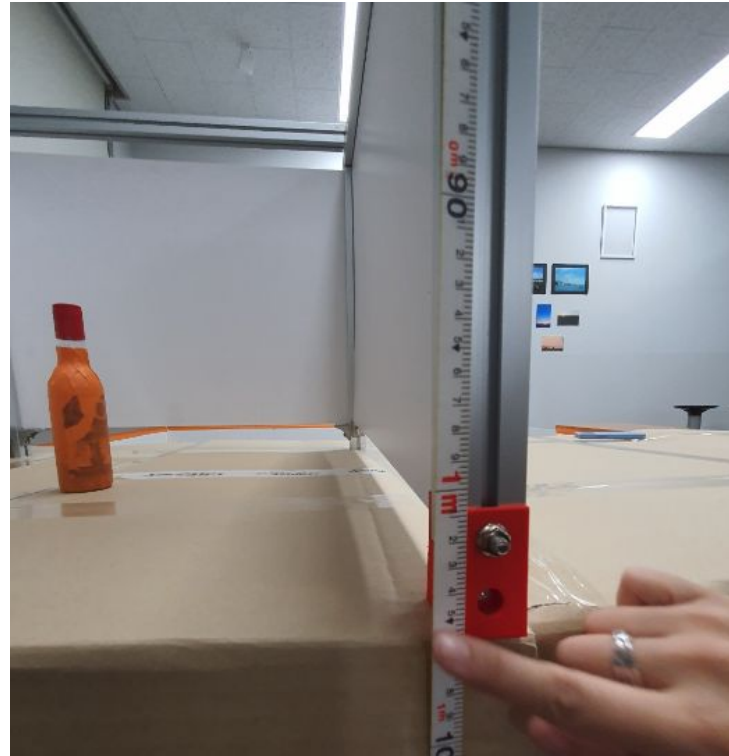
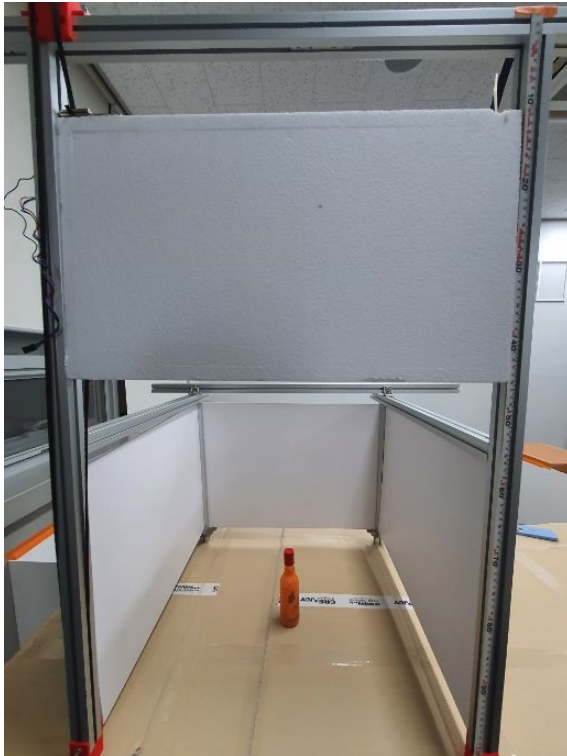
- The center chamber wall panels are of approximately $0.6\text{m} \times 0.6\text{m} \times 0.45\text{m}$
- One of the wall is motorized and it moves vertically to open the door (Dynamic obstacle)
- For Task 3, the door will stay opened all the time.
- For Task 4, the door is closed when start and you need to open it by completing a quest.



The chamber plan w/o overhang

Center chamber (cont)

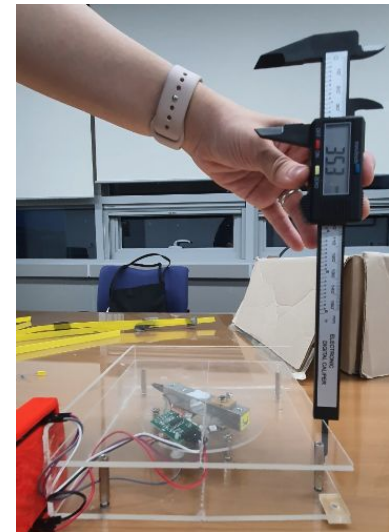
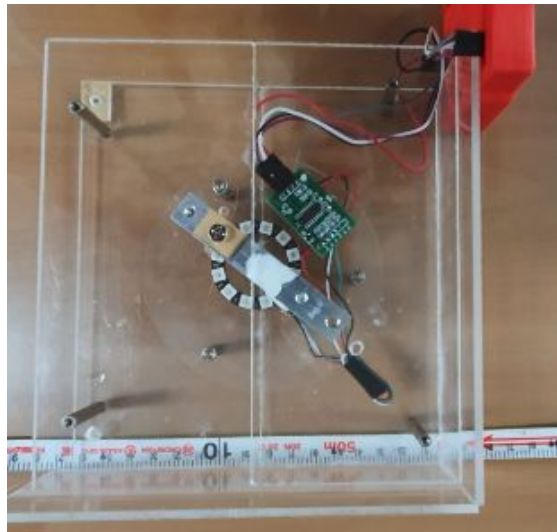
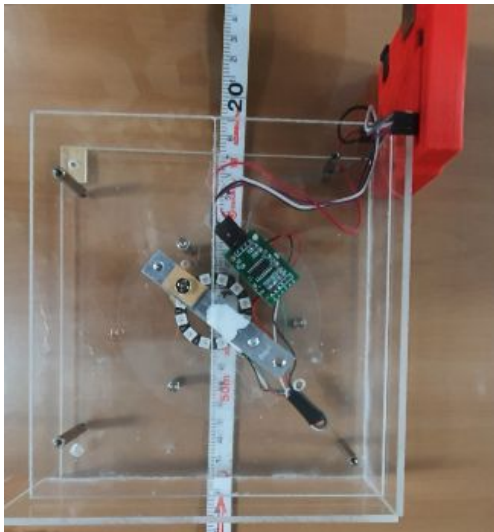
- Please note that there is some overhang of aluminium towards the back of the chamber
- Expect about 17mm of extrusion from the wall panels from 0.5m from the ground.
- The gate is about 35mm tall and will be about 600mm above the ground when opened.
- Chamber is fragile - Designed mainly as line of sight blocker. Please take care when moving around.



The chamber dimension

Pressure pad

- A pressure pad (Figure) will be used for Task 2 and 4 and be placed at top left corner.
- Once it has sensed enough weight, the LED will be turned on. Additionally for Task 4, the door will be activated for.
- The purpose is to test your robot's ability
 - to deal with z offsets (height) when picking up and placing down the bottle with the manipulator arm
 - to deal with dynamic obstacles in the environment.



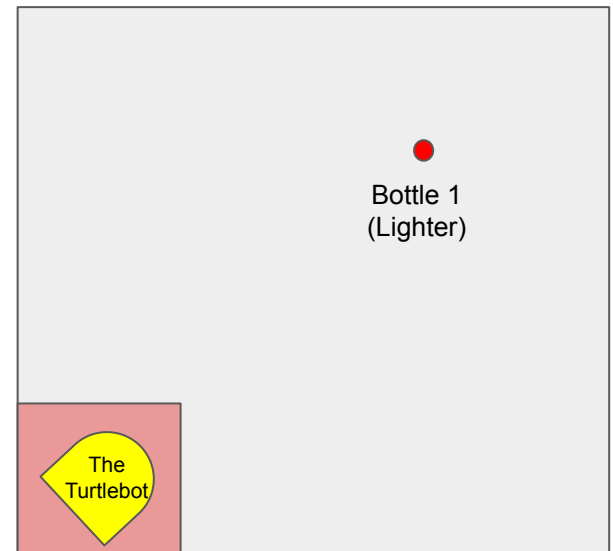
The Pressure pad
(170mm *170mm *35.3mm)

Scoring Rules

- In each task, there is a list of subtasks. We calculate the task score based on *subtasks* (see pp.12-15 for subtasks).
- For each task, each team can attempt maximum five times.
 - Among the five attempts, the one with the highest score is considered as the task score.
 - For each task, **top 3** teams get additional **1.5 pts** in addition to their original task score.
 - The top 3 is decided based on the task score. If there's a tie, the team completed the task in shorter time gets the higher rank.
 - Thus, a task score can exceed 7.5.
- The *final score* of your program is the *sum* of four task scores.
- The maximum value of overall project score is 30. If a team's score exceeds 30, we cap the score (i.e., a project score cannot exceed 30).

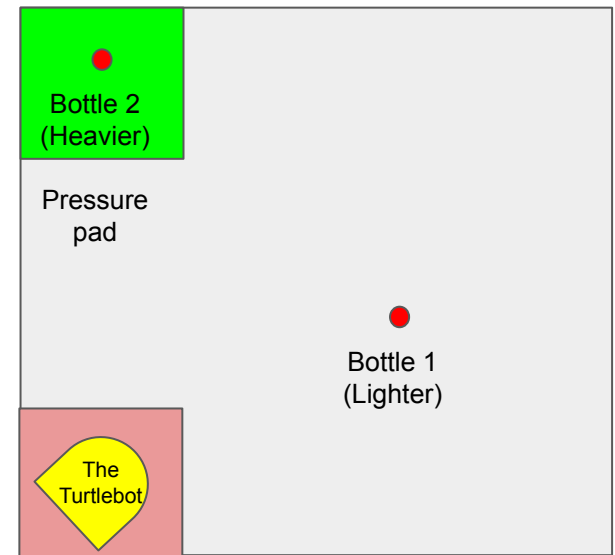
Task 1's Specification (7.5 pts)

- Goal: Approach a bottle and pick it up
- Setting:
 - » Initially, the turtlebot is in the base, and it is looking at the center of the arena.
 - » The bottle is the only object in the arena. It is located **near** the opposite side of the arena.
 - » Although the bottle is not at the center of the field of view of the Turtlebot, we guarantee that the bottle will be within the field of view of the Turtlebot at the beginning.
- Checklist of the subtasks:
 - » (0.75 pts) The turtlebot can locate and look straight at the bottle.
 - » (1.5 pts) The turtlebot can move toward the bottle.
 - » (1.5 pts) The turtlebot can stop in front of the bottle, at a distance such that the robot arm can reach the bottle.
 - » (0.75 pts) The robot arm can be lowered and touch the bottle.
 - » (1.5 pts) The robot arm can grasp the bottle.
 - » (1.5 pts) The robot arm can lift the bottle and hold it firmly in mid air.
- Additional rules:
 - » The bottle must be held **at least 20 cm above ground for at least 5 seconds**.



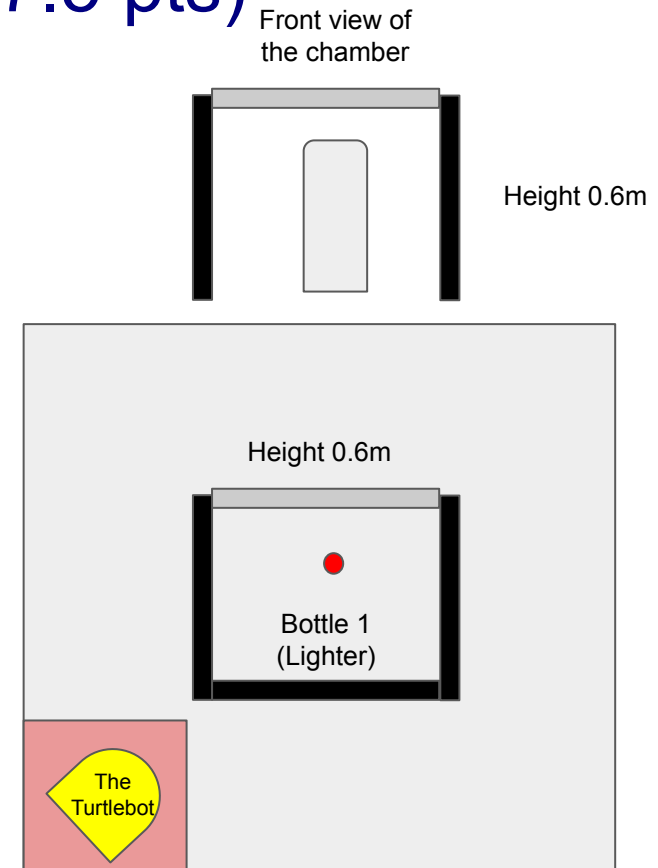
Task 2's Specification (7.5 pts)

- Goal: Bring all bottles back to the Base in an arena without walls
- Setting:
 - Initially, the turtlebot is in the base, and it is looking at the center of the arena.
 - Two bottles are put at **fixed** locations but at different heights
 - No guarantee that the bottles are in the field of view.
 - Bottle 2 will be on a pressure pad (green) while Bottle 1 will be on the floor.
- Checklist of the subtasks:
 - (2.25 pts) The turtlebot can locate and pick up at least one of the bottles.
 - (2.25 pts) The turtlebot can return to the base with at least one of the bottles.
 - (2.25 pts) The turtlebot can retrieve all 2 bottles.
 - (0.75 pts) The turtlebot doesn't collide with the wall or knock down any bottles.
- Additional rules:
 - The turtlebot is considered returned to the base as long as a part of it (e.g., the gripper) is inside the base.
 - A bottle has to stay inside the base after dropping in order to consider the dropping is successfully.



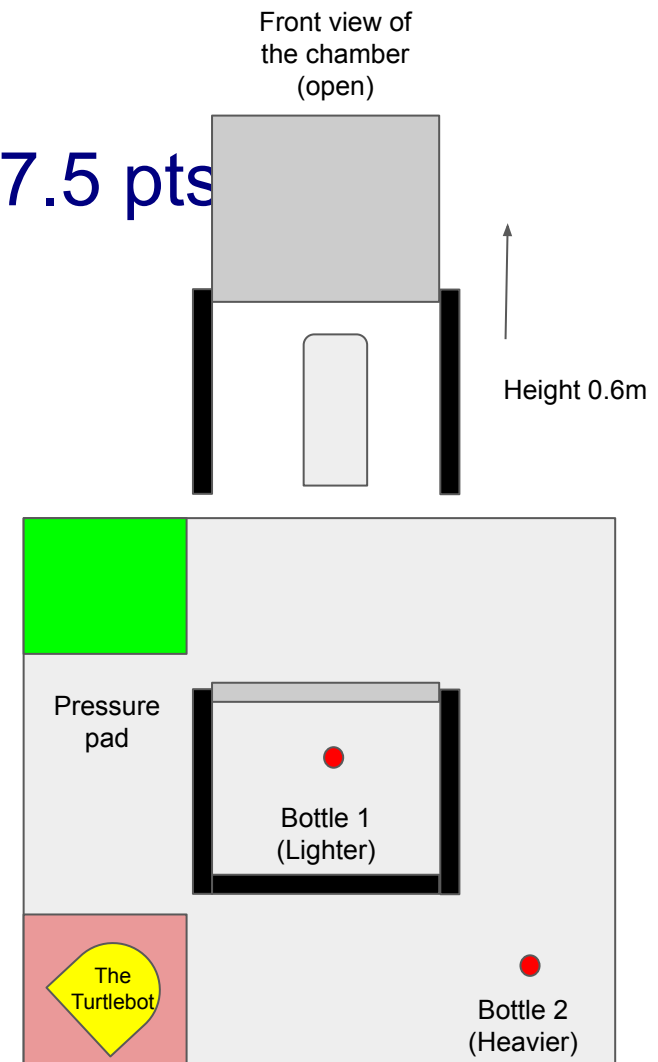
Task 3's Specification (7.5 pts)

- Goal: Remove a bottle from the chamber at the center of the arena
- Setting:
 - » Initially, the turtlebot is in the base, and it is looking at the center of the arena.
 - » One bottle is at a **random** location in the U zone.
 - » There is a U Zone (black) with three walls, the door (gray) will be stay opened.
 - The dimension of the chamber is 0.6 meters x 0.6 meters
 - The chamber will be at the center as shown in the figure.
 - There is height restriction of 0.6m to enter the chamber.
- Checklist of the subtasks:
 - » (2.25 pts) The turtlebot can navigate to the front of the chamber's doors
 - » (2.25 pts) The turtlebot can locate and grab the bottle
 - » (0.75 pts) The turtlebot can remove the bottle from the chamber
 - » (2.25 pts) The turtlebot doesn't collide with the wall or knock down any bottles in the process.
- Additional rules:
 - » The actuation mechanism of the chamber's door will be secured in task 3 so that the door do not fall down.
 - » The bottle is considered in the chamber as long as a part of it is inside the chamber.



Task 4's Specification (7.5 pts)

- Goal: Remove bottle 2 from the chamber after unlocking it.
- Setting:
 - » Initially, the turtlebot is in the base, and it is looking at the center of the arena.
 - » Two bottles are at **random** locations in the arena.
 - No guarantee that the bottles are in the field of view.
 - However, approximate locations are provided in the map on the right
 - » The chamber is originally locked, but the door can be vertically raised by placing Bottle 2 on the pressure pad (green)
 - The bottle must stay on top of the pressure pad to stay the door opened. If pressure is removed the door will be closed.
 - Consider some time for the door is mechanically opened.
- Checklist of the subtasks:
 - » (2.25 pts) The turtlebot can locate and pick up the Bottle 2
 - » (2.25 pts) The turtlebot can open the chamber for at least 3 seconds
 - » (1.5 pts) The turtlebot can remove the Bottle 1 from the chamber
 - » (1.5 pts) The turtlebot doesn't collide with the wall or knock down any bottles.
- Additional rules:
 - » The actuation mechanism of the chamber's door will be motorized in task 4.
 - » You can retry if actuation mechanism of the chamber's door fails. (Refunded trial count). We will manually open the door if issue persists.
 - » A bottle is considered in the chamber as long as a part of it is inside the chamber.



Additional ground rules

- You can choose to either use battery or be plugged in during evaluation
 - No additional points for being plugged in, however.
 - When battery goes low, we will stop the attempt and swap in a new battery. Your trial count is refunded but you have to start over the trial.
- Either Professor Lee or TA reserves right to ask you to stop the robot during evaluation
 - This is to protect robots' motors from being damaged if your robot tries to do impossible commands (e.g. trying to go through a physical wall for more than few seconds)
 - Your trial will not be refunded in this case.
- Your evaluation will end if you exhaust 5 attempt counts for each tasks or reach 2 hour limit
- A little bit of coding for adjustment is allowed between attempts or tasks

Schedule

Evaluation Day Application									
https://doodle.com/meeting/organize/id/e0R4P9Vb									
	June 2022								
	Tue 21				Wed 22				
	01:00 PM - 03:00 PM	03:00 PM - 05:00 PM	05:00 PM - 07:00 PM	08:00 PM - 10:00 PM	10:00 AM - 12:00 PM	01:00 PM - 03:00 PM	03:00 PM - 05:00 PM	05:00 PM - 07:00 PM	08:00 PM - 10:00 PM
Docker	YES								
Robocats		YES							
TaeYeong Kim (Team WafflePi)			YES						
TBD2				YES					
Doyeon Kim					YES				
SHU						YES			
김강연							YES		
m1-wrestlers								YES	
NoX86									

- It is a hard schedule and we don't allow extensions. Use given time slot efficiently.
- Right after your evaluation, please return the two SD cards that will be reused in the future robotics class.
- Please submit your code on BB by the end of the evaluation day

PostScript: Mechanised door opening video (Task 4)

- Note: The door only checks if pressure pad is activated **after it stops**. This is demonstrated in the video. Expect about 9-13s to open the gate fully.

(https://drive.google.com/file/d/1Eaxael_tZWpewolT7yyD8moQJDmmeCB0/view?usp=sharing)

